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Machine Learning-Driven Analysis for Fetal Health Classification Using CTG Data- A Survey

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ABSTRACT

Fetal health monitoring is one of the complicated and critical process during pregnancy and labour. It is also critical for reducing infant mortality, which remains a global concern with around 2.3 million of those deaths occurring within the first month of life, signifying portion of under-one child mortality. The labour also causes maternal mortality. Cardiotocograms (CTGs) are essential tools that evaluate fetal health through ultrasound-based monitoring of Fetal Heart Rate (FHR) and Uterine Contractions (UC). Interpreting CTG results can be time-consuming and difficult, especially in underdeveloped regions that lack access to expert obstetricians.

The research evaluates multiple CTG-derived elements which help forecast fetal health through an analysis utilizing SVM, RF, MLP and KNN algorithms. The analysis confirmed that different attributes exert influence on fetal health through regression and correlation techniques. Moreover, RF has consistently shown better results than other algorithms when it comes to key performance metrics like accuracy, support, precision, and recall F1-score. This makes it a standout choice for our needs.

The study introduces that the attributes help to classify if the fetal health is Normal, Suspicious or Pathological. On comparing different algorithms resulted in accuracy levels ranging between 89-99% when used for fetus health prediction.

The analysis showed that machine learning algorithms succeed in fetal health classification using CTG data while defining the leading features that drive these assessments. This

improved accuracy and clarity can enhance CTG interpretation, making it more efficient and accessible, ultimately benefiting maternal and fetal health outcomes. By using such technologies in clinical settings can be particularly transformative, especially in resource-limited areas, by ensuring timely interventions and contributing to the global effort to reduce preventable child mortality.

KEYWORDS: Fetal health, Classification, Cardiotocography, Machine Learning, Fetal Heart Rate.

INTRODUCTION

Female life becomes its happiest during the time of pregnancy. A pregnancy mandates women to prioritize their health since the fetus depends on her body. The monitoring of fetal development and growth depends on multiple suggested tests, for the mother after every 3 months, known as a trimester. One of those tests is known as Cardiotocogram (CTG). Another name for CTG is Electronic Fetal Monitoring (EFM). Transducers to measure fetal heart rate (FHR) is positioned on the abdomen skin of a pregnant. The recording of FHR simultaneously is obtained from cardiotocogram (CTG). It captures observation the pregnant woman's constrictions and other types of fetal monitoring as well [20]. The obstetrician employs the tool called a cardiotocograph to track the fetal heartbeat along with uterine contractions during prenatal and childbirth periods.

Contractions of the uterus in labor reduce maternal perfusion and oxygenation of the developing fetus, leading to transient hypoxia. Although the majority of babies are physiologically resilient to tolerate intrapartum hypoxia, some whose mothers are subjecting them to extreme hypoxia or have limited physiological reserves could suffer from neurological or fatal injury during labor. Cardiotocography (CTG) monitoring was invented to help detect babies at risk of developing hypoxia through recognition of alterations in fetal heart rate (FHR) pattern. CTG monitoring is in common usage during intrapartum care for the identification of fetal hypoxia, but clinical usefulness is compromised by a relatively poor positive predictive value (PPV) of an abnormal CTG and substantial inter and intra observer variability in interpretation of CTG. [12]

A cardiotocograph serves as the machine which conducts monitoring operations. FHR represents the number of heartbeat per minute (BPM) [3]. By the use of the technique, doctors checks whether the fetus is healthy before and during labour.

- Perinatal mortality describes the death of a fetus between 22nd weeks of pregnancy through birth together with seven days post-birth period. WHO reports perinatal deaths total about 6.3 million annually. The report evaluates perinatal cases by representing them as values per 1000 births alongside considering national development levels as key determinants of these rates. Developed nations maintain a perinatal rate under 10 although developing nations experience a higher figure higher than 30. [4]

Ensuring well-being of the fetal during pregnancy and labor is a critical aspects of obstetric care. In 2020, an estimated 800 women per day died every day from pregnancy and childbirth complications that were completely avoidable. Nearly 95% of maternal deaths occurring in 2020 were from women residing in low income and lower middle-income nations. Early identification of fetal distress can greatly improve clinical decision-making and reduce the risks of complications during childbirth. This technique was introduced in the Sixties in order to avoid foetal death and/or neonatal cerebral palsy; indeed, it has allowed a significant reduction in perinatal mortality and morbidity. Almost 95% Cardiotocography (CTG), a non-invasive monitoring technique, plays a vital role in this process by recording fetal heart rate (FHR) and uterine contractions (UC). This method was brought in the Sixties as a measure against foetal death and/or cerebral palsy neonatal; indeed, it has facilitated a very large reduction of perinatal morbidity and mortality [9]. However, interpreting CTG traces manually is a subject to inter-observer variability, inconsistency, and potential delays, especially in high-stress or resource-constrained environments. While CTG is associated with a high rate of false positives, it is an important tool to monitor circumstances that cause fetal distress, like prolonged spontaneous rupture of membranes, prematurity, and intrauterine growth re-striction [14]. The simultaneous monitoring of the maternal heart rate (MHR) has its utility in some maternal pathological conditions and in situations where it is not possible to differentiate between the mother's and the foetus's heart rates (e.g., complete fetal heart block). Some of the CTG machines offer the option of ongoing MHR monitoring. [3]

The aim of Fetal Health Classification is to produce an automated classification system that evaluates CTG data to place fetuses into Normal, Suspicious and Pathological groups. The system applies modern machine learning techniques to assist medical staff in fast and precise clinical choices resulting in lower infant mortality numbers along with decreased maternal injuries.

This project is not only relevant but also necessary due to the following reasons:

1. Clinical Challenges: Manual interpretation of CTG is prone to errors due to its complexity and dependency on the expertise of healthcare providers.
2. Global Health Concerns: Complications during labor remain a leading cause of perinatal mortality and morbidity worldwide, particularly in low-resource settings.
3. Technological Opportunities: Advances in machine learning provide a unique opportunity to automate and standardize the analysis of CTG, making it more accessible and reliable.

This fetal health classifier model assists obstetricians and midwives in interpreting CTG data more accurately and quickly while reducing errors caused by human variability in CTG interpretation. It provides a tool that can be implemented in low-resource settings where expert analysis may not be readily available. It also serves as a training aid for medical students and professionals to understand fetal health patterns.

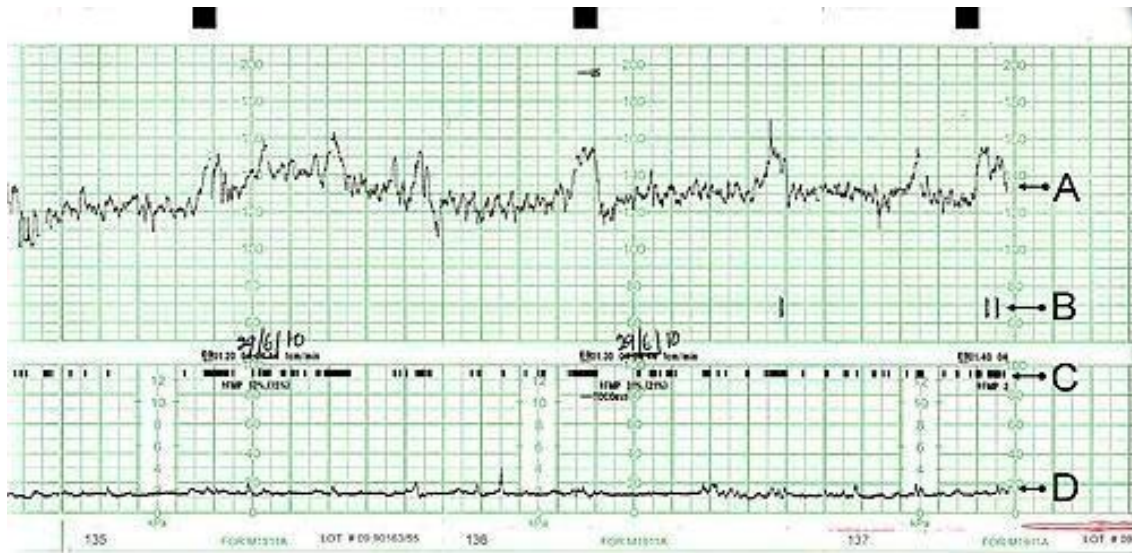


Figure 1: A typical CTG output for a woman not in labour. A: An ultrasound displays a Fetal heartbeat; B: Indicator shows maternal-identified fetal movements felt by mother (triggered by pressing a button); C: Fetal movement; D: Uterine contractions [5]

LITERATURE REVIEW

The categorisation of foetal health has been carried out in several studies, and foetal health has been a significant area of public health study. More than 30 years after CTG was first discovered and used to predict the likelihood of a foetus being conceived. Four essential aspects are evaluated by cardiotocography (CTG) to monitor foetal health and identify indications of distress: baseline foetal heart rate, variability, accelerations, and decelerations. The MMR and high labour complications can be lowered by using machine learning techniques that aid medical professionals during early judgements in difficult scenarios, such as diagnosis. Foetal health stage categorisation is a difficult problem, but machine learning classification approaches can handle it quite well. SVM, Neural Networks (NN) and Random Forest are a few of the common classification techniques. The majority of study was devoted to accurately classifying foetal health.

In 2022 Fasihi et al. Using a 1D convolutional neural network (CNN) conducted fetal health classification into three patterns and reached an accuracy level of 91%. The study used new network configuration for their 1D CNN framework. The researchers conducted five clinical dataset tests on their network architecture and discovered the best performance resulted from their 1D CNN design [6].

Piri et al. A research team implemented MOGA to explore various causes behind foetal death. The researchers established MOGA-CD as an approach to gather optimal fetal wellness information. The evaluation showed that DT and RF and GNB and SVM and XGBoost delivered the maximum accuracy for foetal health categorization among minimal data samples [7]

Nandipati et al. evaluated classification models with feature selection for CTG using CARET and Scikit Learn. The study indicated that while Random Forest (RF) produced the best results, naive Bayes, using Python-ML approaches with decreased attributions, achieved an accuracy rate of 91.88% to 100% [8].

In 2022, The researchers at Mehbodniya et al. applied ML algorithms to foetal health prediction through Random Forest (RF), K-nearest neighbours (KNN), Support Vector Machine (SVM), and Multilayer Perceptron (MLP). The classification research demonstrated GB outperformed every other classification system examined in this work [9].

In 2018, Uzun et al. The CTG dataset received analysis using ELM method, worked on its classification process. The classification criteria built their foundation based on the ASUSP (a morphology model). The PCA-14 ELM machine learning algorithm yielded their best experimental outcome of 84.3% accuracy in their research. [10].

Team Ramla [11] used a cardiotocography decision tree classifier served to monitor foetal health. The major goal of this work is to minimise foetal and maternal mortality. The dataset used was 2126 CTG recordings from the UCI repository. Gini index used by Decision Tree which is using Information gain were used. The authors achieved accuracies of 90.12% and 88.87%, highlighting the urgent need for immediate attention to fetal and maternal mortality.

Sharma and his team[12] proposed utilising machine learning to classify foetal health based on cardiotocography data. The worldwide overbearing health issue of perinatal mortality requires immediate action because it affects so many populations. In the medical field, monitoring foetal growth during pregnancy is the most complex and complicated procedure. This article uses mathematical algorithms to provide insights about the status of foetal health, allowing clinicians to provide drugs to both the mother and the foetus. The study used data from the University of Porto, which includes uterine contractions (UC) and foetal heart rate signals (FHR). Using SVM, RF, and KNN, the authors of this article achieved 93%, 94.5%, and 92.53% accuracy, respectively.

Currently, obstetricians manually analyse CTG data, which might lead to errors and dangers [13]. Machine learning (ML) models hold considerable promise for classifying fetal health using cardiotocography (CTG) data, yet several obstacles must be overcome for successful implementation. Data-related challenges involve concerns about data quality and diversity, as CTG data may be noisy, incomplete, and significantly different across pregnancies, which affects model precision and applicability. Furthermore, large, high-quality datasets with precise fetal health annotations are frequently limited and expensive to acquire. Collaborative research across multiple centres could help address these data challenges by supplying a more varied dataset. Clinical and ethical limitations encompass the likelihood of false positives or negatives, possibly leading to unwarranted interventions or overlooked treatments. Ethical issues, like data privacy and potential model misuse, require thorough examination. Furthermore, clinician acceptance and the incorporation of ML models into current workflows demand focus. To increase the accuracy and efficiency of foetal health evaluations, sophisticated and automated approaches should be explored due to limitations.

This review has explored the available literature on the classification of fetal health using various machine learning models, demonstrating how advancements in this field could greatly enhance clinical outcomes. Our investigation has provided valuable insights and limitations that can contribute to a deeper understanding of fetal wellbeing and highlight the potential for more accurate, personalized, and timely interventions in prenatal care. These developments have the potential to significantly impact both maternal and fetal health by enabling better decision-making and early detection of risks.

DISCUSSION

In 2021, the Nabillah et al. used a publicly available dataset taken from the University of California Irvine Machine Learning Repository. They used the Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) models for their analysis, which yield an accuracy rate between 9% and 37% in each scenario used. On the other hand, Extreme Gradient Boosting (XGB), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Light Gradient Boosting (LGBM), and Random Forest (RF) achieve a much greater level of accuracy between 89% and 99% in the same scenarios.

In 2023, Yiqiao et.al discussed a cardiotocography dataset that was gathered from a research paper. This dataset includes CTG measurements collected from various locations worldwide and was used to reach an accuracy rate of 98.79% Extreme Gradient Boosting often referred to as XGB. The Light Gradient Boosting Machine (LGBM) achieved an accuracy of 99.19% while Support Vector Machines (SVM) achieved an accuracy rate of 99.59%.

Yalamanchili et al. worked with six well-known machine learning algorithms for the recurrent classification of fetal stages using the cardiotocography dataset. Using the dataset an accuracy of 85% was achieved by Decision Tree (DT), 93% accuracy by Random Forest (RF), 90% accuracy by K-Nearest Neighbour (KNN), 90% accuracy by Gradient Boosting Classifier (GBC), 85% and 81% accuracy were achieved by Logistic Regression and Support Vector Machine Classifier respectively.

Sahana et al. utilized a database from the Czech Technical University and University Hospital of Brno (CTU-UHB), which comprises 552 intrapartum records collected using the OB TraceVue System. This dataset was used to determine the sensitivity and specificity of various classifiers, including Random Forest, Multilayer Perceptron, Support Vector Machine, and bagging techniques. The researchers were able to achieve a sensitivity and specificity range of 92.7%-96.4% and 92.8%-98.4%, respectively, for these different classification methods.

In 2023, Adem et.al used a dataset with 2126 CTG records that were automatically processed and were utilized to build an ensemble model comprising Random Forest (RF), Gradient Boosting (GBC) and Extreme Gradient Boosting (XGB) to report an accuracy of more than 99%.

In this paper, K. Arun et al. aimed to develop an information system for fetal health classification, based on CNN and Hybrid-CNN with PCA Dimensionality Reduction with the dataset containing 2126 FHR recordings obtained from 2126 pregnant women using CTG Machines. They were able to achieve an accuracy of 78% with the algorithm of CNN and nearly 95.5% accuracy with the Hybrid CNN with PCA.

In their paper, Arokia et al. sought to classify the health of fetuses using cardiotocographic data and various machine learning models. Their results showed that the ensemble algorithm of Random Forest outperformed the other models, while the Support Vector Machine achieved the second-highest accuracy of 93%, trailing behind the RF model.

In 2021, Naveen et.al employed 4 different machine learning models - Logistic Regression, Random Forest, Support Vector Machines, and Naive Bayes - to analyze a fetal health stroke dataset that has been utilized in numerous previous researches. The results of their analysis showed that these models achieved impressive accuracy rates of 99.5%, 98%, 96.5%, and 97%, respectively.

Zafar et al. report the classification of fetal heart rates and a comparative analysis on the basis of cardiotocography with various classifiers, where Naïve Bayes stands out by achieving an accuracy of 83.06% on the UCI machine learning repository dataset.

Rohit et.al used the dataset of 2130 records to achieve an accuracy of 80.52% by Logistic Regression. 88.97% and 85.45% accuracy levels have been achieved by kNN model and SVM model respectively and the highest accuracy of 93.82% is attained by the extra trees model.

Author	Method(s)	Accuracy(H)
Nabillah et.al	ANN, LSTM, XGB, SVM, KNN, LGBM, RF	89 – 99%
Yiqiao et.al	XGB, LGBM, SVM	99.59%
Yalamanchili et.al	DT, RF, KNN, GBC, LR, SVM	93%
Sahana et.al	RF, MLP, SVM	92.8 – 98.4%
Adem et.al	RF, GBC, XGB	> 99%
K.Arun et.al	CNN, CNN-PCA	95.5%
Arokia et.al	SVM, RF, MLP, KNN, RA, CA	> 93%
Naveen et.al	LR, RF, SVM, NB	99.5%
Zafar et.al	NB	83.06%
Rohit et.al	LR, KNN, ET	93.82%

Table 1: Comparison of Methods and Highest Accuracy recorded

Through researches, Artificial Neural Networks and Long Short-Term Memory showed accuracy ranging from 9 - 37%, these low results suggest issues with hyperparameter turning or preprocessing. Where LSTM's potential for temporal data analysis could be harnessed with better optimization.

Another model was a Tree-Based model, whose algorithms achieved high accuracies. Random Forest (RF) achieved 93 - 98%, leveraging ensemble methods to minimize overfitting. Advanced algorithms like Extreme Gradient Boosting and Light Gradient

Boosting Machine reached impressive accuracy levels of 98% to 99%. In contrast, the Decision Tree (DT) method which is simpler achieved just 85% accuracy.

Support Vector Machines (SVM) showed versatility with accuracies ranging from 81% to 99%, by excelling in the management of high-dimensional data. K-Nearest Neighbours (KNN) achieved moderate results, with accuracies between 88% and 90%, but struggled with high-dimensional datasets.

Logistic Regression (LR) proved to be highly effective, by achieving impressive accuracy rates of up to 99.5% on linear datasets, although it encountered challenges with non-linear data. Naïve Bayes (NB) consistently delivered accuracies between 83% and 97%. Its simplicity is an advantage, but the performance is clearly constrained by its assumptions of independence. Convolutional Neural Networks (CNN) exhibited substantial improvements, with accuracy skyrocketing from 78% to 95.5% when employing Principal Component Analysis (PCA). This underscores the critical importance of strategic feature selection in achieving superior results.

It has been observed that ensemble methods like RF, XGB, LGBM have consistently achieved the highest accuracies by combining multiple algorithms together, and the algorithms of Tree-Based Models were the most reliable, often out-performing others. The quality of the dataset is also essential as well-structured datasets and effective preprocessing significantly improves model performance.

Last but not the least dimensionality reduction is also beneficial as techniques like PCA improves model performance, especially in the case of CNNs.

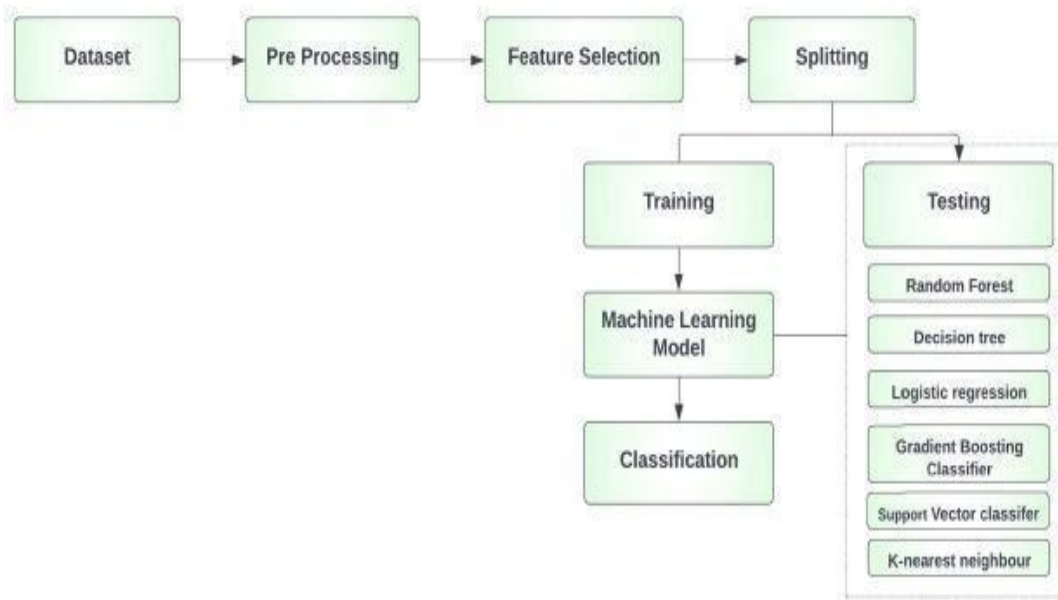


Figure 2: Comprises a Step-by-Step Process from Data Preparation to Model Evaluation.

The usual workflow of a machine learning classification process is depicted in the diagram above. The dataset collection is followed by preprocessing, or cleaning and converting the data so as to fit it into the machine-learning model properly. The most pertinent factors that affect predictions are then selected via feature selection, which lowers dimensionality and

boosts performance. Next, the data is split into two parts: for Training and Testing. The machine learning model learns from the training dataset and its performance is validated on the testing dataset. The model's performance can be verified by employing a range of classification methods like Random Forest, Support Vector Machine (SVM), Decision Tree, Logistic Regression, Gradient Boosting Classifier, and K-Nearest Neighbors.

CONCLUSION

The experiment showcases the ability of machine learning models, especially ensemble methods like Random Forest (RF), Extreme Gradient Boosting (XGB) and LightGBM (LGBM) to effectively classify fetal health using information from cardiotocography. The models mentioned above consistently outperform other methods in terms of accuracy, precision, recall and F1-score, achieving impressive performance rates of 89% or even higher than 99%. Dimensionality reduction methods including Principal Component Analysis (PCA) refine the functionality of advanced models like Convolutional Neural Networks (CNNs). The work, in addition, showcases the capability and power of machine learning application in clinical practice in Low Resource Areas, offering a safe, automated, and reliable CTG interpretation.

Advancements such as these may have vast potential in reducing maternal and fetal mortality by offering immediate and accurate medical intervention. However, the study also highlights the need for high-quality datasets and preprocessing in order to obtain optimal model performance. Challenges such as data quality, ethical considerations, and clinician acceptance should be addressed in future efforts to fully integrate these technologies into routine obstetric care. According to algorithms, future research should be focused on hybrid models, such as CNN-RF combinations, to optimise feature extraction and decision making. Explainable AI (XAI) is essential for increasing model interpretability in healthcare, enabling clinicians to trust. Addressing data imbalance through methods like oversampling or SMOTE can improve model reliability. Prenatal care can be transformed by streaming data monitors by developing Real-time systems. Cross-dataset validation is essential to guarantee that the model is generalizable and robust. Also, the inclusion of domain knowledge, for instance, obstetric expertise, into machine learning algorithms can greatly enhance their accuracy and relevance for specific applications.

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